

The General-Matrix Quantum Boolean Radius: Exact Collapse and Homogeneous Survival

Automated mathematics run `fa_banach.001`, lane 12

August 9, 2026

Status. Candidate full class-by-class answer, likely valid. Under the literal extension of quantum Boolean radius from self-adjoint to arbitrary matrices, the radius of the all-matrix and degree-at-most- d classes is zero. For every positive homogeneous degree, the arbitrary-matrix radius remains within the factor $(2/\pi)^{1/d}$ of the self-adjoint radius. Thus the source theory degenerates exactly in the two classes that mix degree zero with positive degree, while its homogeneous asymptotic orders survive.

1 Source problem

Volberg and Zhang study the Pauli expansion

$$A = \sum_{\mathbf{s} \in \{0,1,2,3\}^n} \hat{A}_{\mathbf{s}} \sigma_{\mathbf{s}}, \quad |\mathbf{s}| = \#\{j : s_j \neq 0\}, \quad (1)$$

in $M_2(\mathbb{C})^{\otimes n}$ and define the quantum Boolean radius of a self-adjoint nonscalar matrix by

$$\sum_{\mathbf{s}} |\hat{A}_{\mathbf{s}}| \text{qBr}_n(A)^{|\mathbf{s}|} = \|A\|. \quad (2)$$

Their Theorem 5.5 compares the resulting radii for four classes—all matrices, homogeneous matrices, exactly d -homogeneous matrices, and degree-at-most- d matrices—with the corresponding real Boolean radii [1]. Remark 5.4 on page 16 says:

“We leave the problem for general A to future study.”

We answer the literal general-matrix version of those same four classes. For clarity, scalar matrices are omitted when taking class infima: their radius is not uniquely specified by (2), and they cannot lower any of the infima below.

2 Definitions

For a nonscalar arbitrary matrix A with expansion (1), put

$$M_A(r) = \sum_{\mathbf{s}} |\hat{A}_{\mathbf{s}}| r^{|\mathbf{s}|}.$$

The normalized trace bound gives $|\hat{A}_{\mathbf{0}}| \leq \|A\|$, with strict inequality unless A is scalar. Hence $M_A(r) = \|A\|$ has a unique positive solution; this is the general-matrix value $\text{qBr}_n^{\mathbb{C}}(A)$. Write

Definition 5.3. The *quantum Boolean radius* of self-adjoint $A \in M_2(\mathbb{C})^{\otimes n}$ is the positive real number $\text{qBr}_n(A)$ such that

$$\sum_{\mathbf{s} \in \{0,1,2,3\}^n} |\widehat{A}_{\mathbf{s}}| \text{qBr}_n(A)^{|\mathbf{s}|} = \|A\|.$$

Similarly, one may define for a class $\mathcal{F}'_n$ of self-adjoint matrices in $M_2(\mathbb{C})^{\otimes n}$

$$\text{qBr}_n(\mathcal{F}'_n) := \inf\{\text{qBr}_n(A) : A \in \mathcal{F}'_n\}.$$

Remark 5.4. In [DMP18] only the real-valued functions were considered and some arguments rely on the real structure. We are not going to discuss the possible extension to the complex-valued functions here. It is for this reason we require A to be self-adjoint. In fact, when A is self-adjoint, the function $f_A = \text{tr}[Ap(\cdot)]$ on $\{-1, 1\}^{3n}$ constructed in Proposition 2.2 is real-valued and has real Fourier coefficients, so that we can use the results in [DMP18] directly. We leave the problem for general A to future study.

If $\mathcal{F}_n$ denotes one of the four classes of functions (1-4) on $\{-1, 1\}^n$ listed as above, then we use $\mathcal{F}_n^q$ to denote the quantum counterpart of class of matrices in $M_2(\mathbb{C})^{\otimes n}$. For example, if $\mathcal{F}_n = \mathcal{F}_n(\leq d)$ is the class of polynomials on $\{-1, 1\}^n$ of degree at most d , then $\mathcal{F}_n^q = \mathcal{F}_n^q(\leq d)$ denotes the class of self-adjoint $A \in M_2(\mathbb{C})^{\otimes n}$ of degree at most d .

Figure 1: Definition 5.3 and Remark 5.4 on page 16 of arXiv:2210.14468v4. The full readable width is retained.

$\text{qBr}_n^{\mathbb{C}}(\mathcal{F})$ for the infimum over arbitrary matrices in a class $\mathcal{F}$, and $\text{qBr}_n^{\text{sa}}(\mathcal{F})$ for the source's self-adjoint infimum.

Likewise, for a nonscalar complex-valued function $f = \sum_{S \subseteq [n]} \widehat{f}(S) \chi_S$ on the Boolean cube, let $\text{Br}_n^{\mathbb{C}}(f)$ solve

$$\sum_{S \subseteq [n]} |\widehat{f}(S)| |\text{Br}_n^{\mathbb{C}}(f)|^{|S|} = \|f\|_{\infty}.$$

Class infima are denoted $\text{Br}_n^{\mathbb{C}}(\mathcal{F})$; the source's real-valued counterparts are denoted $\text{Br}_n^{\mathbb{R}}(\mathcal{F})$.

Let $\mathcal{H}_{n,d}^{\mathbb{C}}$ be the space of matrices whose Pauli support is exactly in degree d , and let $\mathcal{H}_{n,d}^{\text{sa}}$ be its self-adjoint part. We interpret hom_+ as the union of the positive homogeneous sectors $d = 1, \dots, n$. This is the nonconstant part of the source class of homogeneous functions.

3 Idea of the proof

The obstruction is phase cancellation across degrees. A scalar multiple of the identity and an imaginary Pauli string can form a unitary matrix. The operator norm sees their orthogonal phases quadratically, while the Boolean majorant in (2) adds their absolute coefficients. Moving the scalar coefficient toward one forces the radius to zero. This immediately settles the two source classes that permit a mixture of degrees zero and one.

That mechanism is unavailable inside a fixed positive degree. There the radius is simply the reciprocal d th root of an ℓ_1 coefficient-to-norm constant. Rotating an arbitrary matrix by a scalar phase and taking its real part gives a self-adjoint matrix without increasing norm. Averaging over the phase preserves exactly $2/\pi$ of every coefficient's modulus. Thus the arbitrary complex coefficient constant is at most $\pi/2$ times its self-adjoint counterpart. This proves that every homogeneous sector, and therefore their union, retains the source's asymptotic order.

Finally, the source's matrix-to-Boolean reduction applies to arbitrary matrices. It produces a complex-valued function. With the complex Boolean radius, its coefficient identity proves the factor-three comparison verbatim.

4 Main theorem

Theorem 1 (General-matrix quantum Boolean radii). *Let $n \geq 1$.*

(i) *The class of all arbitrary matrices satisfies*

$$\boxed{\text{qBr}_n^{\mathbb{C}}(\text{all}) = 0.}$$

(ii) *For every $1 \leq d \leq n$, the arbitrary degree-at-most- d class satisfies*

$$\boxed{\text{qBr}_n^{\mathbb{C}}(\leq d) = 0.}$$

(iii) *For every $1 \leq d \leq n$,*

$$\boxed{\left(\frac{2}{\pi}\right)^{1/d} \text{qBr}_n^{\text{sa}}(=d) \leq \text{qBr}_n^{\mathbb{C}}(=d) \leq \text{qBr}_n^{\text{sa}}(=d).} \quad (3)$$

(iv) For the union of positive homogeneous degrees,

$$\boxed{\frac{2}{\pi} \text{qBr}_n^{\text{sa}}(\text{hom}_+) \leq \text{qBr}_n^{\mathbb{C}}(\text{hom}_+) \leq \text{qBr}_n^{\text{sa}}(\text{hom}_+).} \quad (4)$$

Proof. Let $P = \sigma_1 \otimes I^{\otimes(n-1)}$. Thus $P = P^*$, $P^2 = I$, and P has Pauli degree one. For $0 < a < 1$, set

$$b = (1 - a^2)^{1/2}, \quad A_a = aI + ibP.$$

The two terms commute and

$$A_a^* A_a = (aI - ibP)(aI + ibP) = (a^2 + b^2)I = I,$$

so A_a is unitary and $\|A_a\| = 1$. Its only nonzero Pauli coefficients are a in degree zero and ib in degree one. Therefore

$$\text{qBr}_n^{\mathbb{C}}(A_a) = \frac{1-a}{b} = \sqrt{\frac{1-a}{1+a}} \rightarrow 0 \quad (a \uparrow 1). \quad (5)$$

Every A_a belongs both to the all-matrix class and to every degree-at-most- d class with $d \geq 1$. Since radii are nonnegative, (5) proves (i) and (ii).

Fix a positive degree d . For $A \in \mathcal{H}_{n,d}^{\mathbb{C}}$, write

$$L_d(A) = \sum_{|\mathbf{s}|=d} |\hat{A}_{\mathbf{s}}|.$$

Homogeneity gives

$$\text{qBr}_n^{\mathbb{C}}(A) = \left(\frac{\|A\|}{L_d(A)} \right)^{1/d}. \quad (6)$$

Define the finite coefficient constants

$$K_{n,d}^{\mathbb{C}} = \sup_{0 \neq A \in \mathcal{H}_{n,d}^{\mathbb{C}}} \frac{L_d(A)}{\|A\|}, \quad K_{n,d}^{\text{sa}} = \sup_{0 \neq A \in \mathcal{H}_{n,d}^{\text{sa}}} \frac{L_d(A)}{\|A\|}.$$

The inclusion of the self-adjoint space gives $K_{n,d}^{\text{sa}} \leq K_{n,d}^{\mathbb{C}}$.

For arbitrary $A \in \mathcal{H}_{n,d}^{\mathbb{C}}$ and $\theta \in [0, 2\pi]$, let

$$H_\theta = \text{Re}(e^{-i\theta} A) = \frac{e^{-i\theta} A + e^{i\theta} A^*}{2}.$$

Each Pauli string is self-adjoint, so H_θ lies in $\mathcal{H}_{n,d}^{\text{sa}}$ and has coefficient $\text{Re}(e^{-i\theta} \hat{A}_{\mathbf{s}})$ at $\mathbf{s}$. Moreover, $\|H_\theta\| \leq \|A\|$. The scalar identity

$$\frac{1}{2\pi} \int_0^{2\pi} |\text{Re}(e^{-i\theta} z)| d\theta = \frac{2}{\pi} |z| \quad (7)$$

therefore yields

$$\begin{aligned} \frac{2}{\pi} L_d(A) &= \frac{1}{2\pi} \int_0^{2\pi} L_d(H_\theta) d\theta \\ &\leq K_{n,d}^{\text{sa}} \|A\|. \end{aligned}$$

Consequently

$$K_{n,d}^{\text{sa}} \leq K_{n,d}^{\mathbb{C}} \leq \frac{\pi}{2} K_{n,d}^{\text{sa}}. \quad (8)$$

Taking reciprocal d th roots in (8), using (6), proves (iii).

Finally, the radius of the positive homogeneous union is the infimum of the exact-degree radii over $1 \leq d \leq n$. Since $(2/\pi)^{1/d} \geq 2/\pi$, taking infima in (iii) proves (iv). $\square$

5 The complex Boolean cube and the full comparison

The same phase obstruction and phase averaging describe the complex-valued analogue on the classical Boolean cube.

Theorem 2 (Complex Boolean radii). *For every $n \geq 1$ and $1 \leq d \leq n$,*

$$\text{Br}_n^{\mathbb{C}}(\text{all}) = 0, \quad \text{Br}_n^{\mathbb{C}}(\leq d) = 0,$$

while

$$\left(\frac{2}{\pi}\right)^{1/d} \text{Br}_n^{\mathbb{R}}(= d) \leq \text{Br}_n^{\mathbb{C}}(= d) \leq \text{Br}_n^{\mathbb{R}}(= d) \quad (9)$$

and

$$\frac{2}{\pi} \text{Br}_n^{\mathbb{R}}(\text{hom}_+) \leq \text{Br}_n^{\mathbb{C}}(\text{hom}_+) \leq \text{Br}_n^{\mathbb{R}}(\text{hom}_+). \quad (10)$$

Proof. For $0 < a < 1$ and $b = (1 - a^2)^{1/2}$, the function $f_a(x) = a + ibx_1$ has $\|f_a\|_{\infty} = 1$ and

$$\text{Br}_n^{\mathbb{C}}(f_a) = \frac{1 - a}{b} = \sqrt{\frac{1 - a}{1 + a}} \rightarrow 0.$$

This proves both zero-radius statements. In exact degree d , define the ℓ_1 Fourier coefficient-to-supremum-norm constants over complex- and real-valued functions. For complex f , the real-valued function $g_{\theta} = \text{Re}(e^{-i\theta} f)$ satisfies $\|g_{\theta}\|_{\infty} \leq \|f\|_{\infty}$. Applying (7) coefficientwise gives the same comparison $K_{n,d}^{\mathbb{R}} \leq K_{n,d}^{\mathbb{C}} \leq (\pi/2) K_{n,d}^{\mathbb{R}}$ as in (8). Reciprocal d th roots prove (9); taking the infimum over positive degrees proves (10). $\square$

The next theorem is the exact arbitrary-matrix replacement for Theorem 5.5 of the source.

Theorem 3 (Complex quantum-classical comparison). *Let $\mathcal{F}$ be any of the four classes all, hom_+ , $(= d)$, or $(\leq d)$, with arbitrary matrices on the quantum side and complex-valued functions on the classical side. Then*

$$\boxed{\text{qBr}_n^{\mathbb{C}}(\mathcal{F}_n^q) \leq \text{Br}_n^{\mathbb{C}}(\mathcal{F}_n), \quad \text{Br}_{3n}^{\mathbb{C}}(\mathcal{F}_{3n}) \leq 3 \text{qBr}_n^{\mathbb{C}}(\mathcal{F}_n^q).} \quad (11)$$

Proof. For complex-valued $f(x) = \sum_{S \subset [n]} a_S \chi_S(x)$, the diagonal matrix

$$A_f = \sum_{S \subset [n]} a_S \prod_{j \in S} \sigma_3^{(j)}$$

has the values $f(x)$ as its diagonal entries. Hence $\|A_f\| = \|f\|_{\infty}$, its Pauli coefficients are the a_S , and it preserves support classes and radius. Taking infima proves the first inequality.

Conversely, Proposition 2.2 of [1] applies to every arbitrary matrix A and produces a complex-valued function f_A on $\{-1, 1\}^{3n}$ with $\|f_A\|_{\infty} \leq \|A\|$, the same degree type, and coefficients

$$\widehat{f}_A(q(\mathbf{s})) = 3^{-|\mathbf{s}|} \widehat{A}_{\mathbf{s}}, \quad \widehat{f}_A(S) = 0 \quad (S \notin \text{im } q).$$

At $r = 3 \text{qBr}_n^{\mathbb{C}}(A)$, the coefficient majorant of f_A is exactly $\|A\|$, which is at least $\|f_A\|_{\infty}$. Monotonicity gives $\text{Br}_{3n}^{\mathbb{C}}(f_A) \leq 3 \text{qBr}_n^{\mathbb{C}}(A)$. Taking infima proves the second inequality. $\square$

6 Consequences for the source theorem

Let $\text{Br}_n^{\mathbb{R}}(=d)$ and $\text{Br}_n^{\mathbb{R}}(\text{hom})$ denote the real Boolean radii used by Volberg and Zhang. Their Theorem 5.5 says, in the two homogeneous classes,

$$\frac{1}{3} \text{Br}_{3n}^{\mathbb{R}}(\mathcal{F}_{3n}) \leq \text{qBr}_n^{\text{sa}}(\mathcal{F}_n^q) \leq \text{Br}_n^{\mathbb{R}}(\mathcal{F}_n).$$

Corollary 4. *For arbitrary matrices and $1 \leq d \leq n$,*

$$\frac{1}{3} \left(\frac{2}{\pi} \right)^{1/d} \text{Br}_{3n}^{\mathbb{R}}(=d) \leq \text{qBr}_n^{\mathbb{C}}(=d) \leq \text{Br}_n^{\mathbb{R}}(=d), \quad (12)$$

and

$$\frac{2}{3\pi} \text{Br}_{3n}^{\mathbb{R}}(\text{hom}) \leq \text{qBr}_n^{\mathbb{C}}(\text{hom}_+) \leq \text{Br}_n^{\mathbb{R}}(\text{hom}). \quad (13)$$

In particular,

$$\text{qBr}_n^{\mathbb{C}}(\text{hom}_+) \asymp \sqrt{\frac{\log n}{n}}.$$

For each fixed d , the exactly d -homogeneous arbitrary-matrix radius has the same order in n as its real and self-adjoint counterparts.

Proof. Combine Theorems 2 and 3. The final orders follow from Theorem 5.2 of [1]. $\square$

Thus there is no positive analogue of the source comparison for the literal general-matrix versions of all and $(\leq d)$: the left-hand quantum radii are exactly zero, while their real/self-adjoint counterparts are positive. Equations (12) and (13) give the surviving general-matrix program.

7 Verification and novelty boundary

The proof is exact and does not depend on computation. The included checker constructs A_a as a matrix for several n and a , verifies unitarity and the formula (5), and numerically checks the phase average (7) for deterministic complex coefficient arrays. Its reported maximum residual is recorded in `verification.md`.

The bounded novelty check was run through August 9, 2026. The four cheap run indexes and web/local sources were searched by arXiv id, title, authors, the quoted sentence, “quantum Boolean radius”, “general matrix”, and “complex-valued Boolean cube”. The search found the source versions, the underlying real theory, and adjacent later work, but no direct statement of the zero-radius obstruction or the homogeneous $\pi/2$ comparison. This was not an exhaustive search.

The result addresses the literal extension of Definition 5.3 and the matching complex-valued Boolean theory. It does not determine the best homogeneous comparison constant or analyze modified definitions that remove constants, phases, or the coefficient majorant.

Human-review recommendation. Promote after checking the literal meaning of “general A ”, the unitary example, and the phase-average passage from (7) to (8). These are the only substantive proof points.

References

- [1] A. Volberg and H. Zhang, *Noncommutative Bohnenblust–Hille inequalities*, Math. Ann. **389** (2024), 1657–1676; arXiv:2210.14468v4.